



BSc (Hons) in **Information Technology**
Specialized in AI

IT3012 : Intelligent Agents

Year 3 · Semester 1 · 2026 Semester 2

Faculty of Computing

Practical 01

Objective & Lecture Mapping: Recall that an intelligent agent acts within a continuous loop: Sensors generate percept sequences, and Actuators execute actions to maximize a Performance measure. Use this sheet to execute practical modifications and incrementally evaluate your design against Lecture 01 and 02 theory (from basic recall to analytical classification).

Part 1: Code Analysis & PEAS Evaluation

Task 0 : Setting up and getting the starter Code

1. Go to the [Git Hub Repo](#). Create a Fork of this Git Repo to your Personal Github Profile. Open it using your favourite IDE and follow the below instruction to complete the practical. The base code is in the "master" brach of the repo.

Task 1.1: The Environment State (`__init__`)

1. (Remember) List the four components of the PEAS framework discussed in Lecture 01.

Performance Measure ,Environment ,Sensors

2. (Understand) Look at the `VisualGridHuntGame.__init__` method. Which specific variables in this code represent the physical "Environment (E)" state?

`self.score`, `self.steps`, and `self.collision` are mainly performance and simulation-status variables rather than the physical layout of the environment

3. (Analyze) Based on the variables initialized (specifically `self.opponents`), classify this baseline environment as "Single-Agent" or "Multi-Agent". Briefly justify your choice.

The variable `self.opponents` indicates that there are other entities (opponents) in the environment besides the main agent/player. Therefore, the environment involves multiple agents interacting within the same environment.

Task 1.2: The Perception Subsystem (`get_percept`)

4. (Remember) Define what a "percept sequence" is according to Lecture 01.

A percept sequence is the complete history of everything an agent has perceived through its sensors from the beginning up to the current moment.

5. (Understand) Which component of the PEAS framework does the `get_percept()` method represent in our code?

`simulator.py`

6. (Evaluate) Based on the exact dictionary returned by `get_percept()` , is this environment "Fully Observable" or "Partially Observable"? Explain why based on what the agent can and cannot see.

The percept dictionary gives the agent its position, whether food is at its current position, the score, and the remaining-food count. It does not provide the locations of all food pellets or walls, so the agent cannot see the complete environment state

Task 1.3: Action Execution (`execute_action`)

7. (Understand) When `execute_action()` deducts points for hitting a wall, which PEAS component is being actively updated?

The Performance measure (P) is being updated.

8. (Evaluate) Why is it structurally important that this metric evaluates changes in the external environment state (hitting a wall) rather than the agent's internal processing effort?

It is important because the Performance measure (P) should evaluate the agent's effect on the environment and whether it achieves its goal.

Part 2: Guided Modification & Deep Theory Mapping

Follow the implementation instructions to inject a new hazard, then answer the questions to map your code changes back to the theoretical nature of the environment.

Step 2.1: Extending the Environment Initialization (`__init__`)

1. **Implementation Instructions:** Declare a new trap collection attribute (e.g., `self.toxic_traps = set()`) inside `__init__`. Populate it with coordinate tuples safely avoiding position `(0, 0)`, walls, and food.

9. (Understand) If you successfully add `self.toxic_traps` to the environment but intentionally **hide** this data from the agent's sensors, how does this specifically alter the environment's "Observability" classification?

It remains Partially Observable, but the agent now has an additional hidden part of the environment state.

Step 2.2: Updating the Perception Subsystem (`get_percept`)

1. **Implementation Instructions:** Locate `get_percept(self)` and add a new boolean sensor key (e.g., `'smells_toxin': tuple(self.agent_pos) in self.toxic_traps`) to the dictionary returned to the agent loop.

10. (Analyze) By adding `'smells_toxin'`, you expand the agent's percept sequence. Explain how this specific sensor helps the agent act with "Rationality" (maximizing expected utility).

smells_toxin improves rational action because it gives the agent information about an immediate hazard before it moves.

Step 2.3: Modifying Action Execution & Visual Rendering (`execute_action` & `draw_grid`)

1. **Implementation Instructions:** In `execute_action`, check if the updated `self.agent_pos` intersects with `self.toxic_traps` to decrement `self.score` by 15 points. In `draw_grid`, iterate through traps to render custom purple shapes on the canvas.

11. (Remember) You programmed a severe score penalty for stepping on a trap to avoid metric exploitation. What was the classic "Vacuum World Exploit" example discussed in Lecture 01 that illustrated the danger of faulty metrics?

The classic Vacuum World exploit was an agent that creates dirt and then cleans it repeatedly to earn reward each time. It achieves a high score by exploiting the metric rather than keeping the environment genuinely clean, showing why

Finalize and Lock Lab 01 Analytical Evaluation



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